

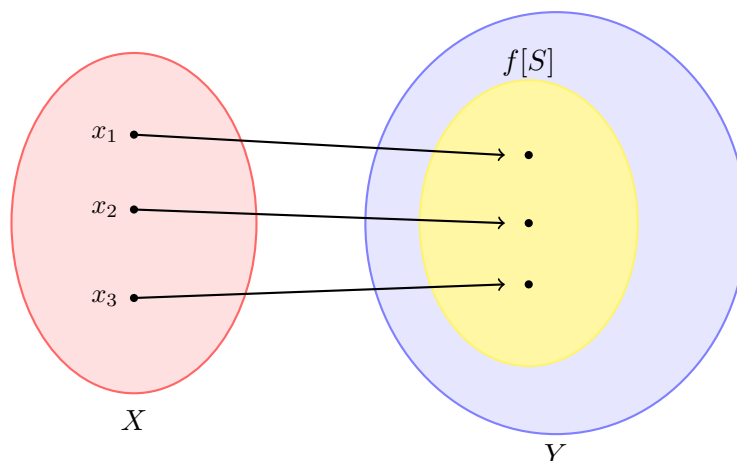
Sets & Functions — Short Notes and Answers

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Recap

- **Set:** a collection of distinct elements. Notation: $x \in A$, $x \notin A$.
- **Subset:** $A \subseteq B \iff (\forall x)(x \in A \Rightarrow x \in B)$; proper subset: $A \subset B$.
- **Common sets:**
 - $\mathbb{N} = \{1, 2, 3, \dots\}$ (*some authors include 0*); $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.
 - $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$.
- **Operations:**
 - Union/intersection: $A \cup B$, $A \cap B$.
 - Difference/complement: $A \setminus B$, A^c (relative to a universe Ω).
 - Symmetric difference: $A \triangle B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.
- **Laws used most:**
 - Commutative & Associative for $\cup, \cap$.
 - Idempotence: $A \cup A = A$, $A \cap A = A$.
 - Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and dually.
 - De Morgan: $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$.
 - Absorption/Identity: $A \cap (A \cup B) = A$, $A \cup (A \cap B) = A$, $A \cup \emptyset = A$, $A \cap \Omega = A$.
- **Notation:**
 - $2 \in \{1, 2, 3\}$ is true, but $\{2\} \in \{1, 2, 3\}$ is false; instead $\{2\} \subseteq \{1, 2, 3\}$.
- **Cartesian product:** $A \times B = \{(a, b) : a \in A, b \in B\}$ (ordered pairs).
- **Binary operation:** $*$: $S \times S \rightarrow S$. Neutral (identity) e : $e * s = s * e = s$ for all s .
 - *Uniqueness:* if e and e' are neutral, then $e = e * e' = e'$.
- **Function:** $f : S \rightarrow T$ assigns each $s \in S$ exactly one $t \in T$.
 - Domain = S , Codomain = T , Image = $f[S] = \{f(s) : s \in S\} \subseteq T$.
 - The word “range” is ambiguous (can mean codomain or image). We will say *codomain* or *image*.



Domain X , codomain Y , image $f[S] \subseteq Y$.

- **Injective:** $f(s_1) = f(s_2) \Rightarrow s_1 = s_2$.
 - **Surjective:** $f[S] = T$.
 - **Bijective:** injective and surjective.
 - **Composition:** $(g \circ f)(x) = g(f(x))$; neutral for composition on $X \rightarrow X$ is id_X .
 - **Inverse:** f has an inverse f^{-1} iff f is bijective; then $f^{-1} \circ f = \text{id}$ and $f \circ f^{-1} = \text{id}$.
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Answers to Questions on the Slides

1. Example of a non-commutative operation

- *Sets:* $(\mathcal{P}(A), \setminus)$ is not commutative. Example: $A = \{1, 2, 3\}$, $X = \{1, 2\}$, $Y = \{2\}$. Then $X \setminus Y = \{1\}$ but $Y \setminus X = \emptyset$.
- *Functions:* composition is usually non-commutative. On $\mathbb{R}$, with $f(x) = x + 1$ and $g(x) = 2x$, we have $(g \circ f)(x) = 2x + 2 \neq 2x + 1 = (f \circ g)(x)$.

2. A pair $(S, *)$ with no neutral element

- $(\mathbb{Z}, -)$ has no neutral. If e were neutral, we would need $s - e = s$ and $e - s = s$ for all s . From $s - e = s$ get $e = 0$; from $e - s = s$ get $e = 2s$ for all s —impossible.

3. Neutral of $(\mathbb{Q}, \times)$ and which elements have inverses

- Neutral: 1.
- Invertible elements: all $q \in \mathbb{Q}$ with $q \neq 0$; inverse is $1/q$. Zero has no multiplicative inverse.

4. Injective/surjective/bijective checks

- $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$:
 - Not injective: $f(a) = f(-a)$ for $a \neq 0$.
 - Not surjective onto $\mathbb{R}$: $f(\mathbb{R}) = [0, \infty)$ (no negatives).
 - Hence not bijective.
- $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x^3$:
 - Injective: strictly increasing; $x_1^3 = x_2^3 \Rightarrow x_1 = x_2$.

- Surjective: for any $y \in \mathbb{R}$, take $x = \sqrt[3]{y}$.
- Hence bijective.

5. **Finite bijection $f : S \rightarrow T$: what must be true?**

- If S, T are finite and f is bijective, then $|S| = |T|$ (one-to-one and onto pairing).

6. **Is that still true for infinite sets?**

- Yes. For any sets, finite or infinite, a bijection means equal cardinality. Example: $\mathbb{N} \leftrightarrow 2\mathbb{N}$ via $n \mapsto 2n$.

7. **Why no inverse for $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^2$?**

- Inverse exists iff bijective. Here x^2 is neither injective (collisions at $\pm a$) nor surjective (no negatives).
- *Fixes to show:* $[0, \infty) \xrightarrow{x^2} [0, \infty)$ is bijective (inverse $\sqrt{\cdot}$). Also, $\mathbb{R} \xrightarrow{x^2} [0, \infty)$ is surjective but not injective.

8. **Neutral element in $(Fun_X, \circ)$**

- $\text{id}_X(x) = x$ is neutral: $f \circ \text{id}_X = \text{id}_X \circ f = f$. Uniqueness: if e and e' are neutral, then $e = e \circ e' = e'$.

9. **Which $f \in Fun_X$ have inverses? How defined?**

- Exactly the bijections $f : X \rightarrow X$. There is a unique $f^{-1} : X \rightarrow X$ with $f^{-1} \circ f = \text{id}_X$ and $f \circ f^{-1} = \text{id}_X$.

10. **Neutral of $(\mathcal{P}(A), \cap)$**

- Neutral is A , since $B \cap A = B$ for all $B \subseteq A$. (In $(\mathcal{P}(A), \cup)$ the neutral is $\emptyset$.)